

Optimization Lecture 10: Equality constrained minimization

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Outline

Equality constrained minimization

Newton's method with equality constraints

Infeasible start Newton method

Implementation

Equality constrained minimization

Equality constrained minimization

- ▶ equality constrained smooth minimization problem:

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & Ax = b\end{array}$$

- ▶ we assume
 - ▶ f convex, twice continuously differentiable
 - ▶ $A \in \mathbf{R}^{p \times n}$ with $\text{rank } A = p$
 - ▶ p^* is finite and attained
- ▶ **optimality conditions:** x^* is optimal if and only if there exists a v^* such that

$$\nabla f(x^*) + A^T v^* = 0, \quad Ax^* = b$$

why? Slater's condition is met. KKT condition is sufficient and necessary.

Equality constrained quadratic minimization

- ▶ $f(x) = (1/2)x^T Px + q^T x + r, P \in \mathbf{S}_+^n$
- ▶ $\nabla f(x) = Px + q$
- ▶ optimality conditions are a **system of linear equations**

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ v^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

- ▶ coefficient matrix is called KKT matrix
- ▶ KKT matrix is nonsingular if and only if

$$Ax = 0, \quad x \neq 0 \implies x^T Px > 0$$

- ▶ equivalent condition for nonsingularity: $P + A^T A > 0$

Noting $P \in \mathbf{S}_+^n$, $x^T Px > 0 \Leftrightarrow Px \neq 0$.

Eliminating equality constraints

- ▶ represent feasible set $\{x \mid Ax = b\}$ as $\{Fz + \hat{x} \mid z \in \mathbf{R}^{n-p}\}$
 - ▶ $\hat{x}$ is (any) **particular solution** of $Ax = b$
 - ▶ range of $F \in \mathbf{R}^{n \times (n-p)}$ is nullspace of A (**rank** $F = n - p$ and $AF = 0$)
- ▶ **reduced or eliminated problem**: minimize $f(Fz + \hat{x})$
- ▶ an unconstrained problem with variable $z \in \mathbf{R}^{n-p}$
- ▶ from solution z^* , obtain x^* and v^* as

$$x^* = Fz^* + \hat{x}, \quad v^* = -(AA^T)^{-1} A \nabla f(x^*)$$

Example: Optimal resource allocation

- ▶ allocate resource amount $x_i \in \mathbf{R}$ to agent i
- ▶ agent i cost = $f_i(x_i)$
- ▶ resource budget is b , so $x_1 + \cdots + x_n = b$
- ▶ resource allocation problem is

$$\begin{array}{ll}\text{minimize} & f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \\ \text{subject to} & x_1 + x_2 + \cdots + x_n = b\end{array}$$

- ▶ eliminate $x_n = b - x_1 - \cdots - x_{n-1}$, i.e., choose

$$\hat{x} = be_n, \quad F = \begin{bmatrix} \mathbf{I}_{(n-1) \times (n-1)} \\ -\mathbf{1}^T \end{bmatrix} \in \mathbf{R}^{n \times (n-1)}$$

- ▶ reduced problem: minimize
 $f_1(x_1) + \cdots + f_{n-1}(x_{n-1}) + f_n(b - x_1 - \cdots - x_{n-1})$

Newton's method with equality constraints

Newton step

- ▶ Newton step Δx_{nt} of f at feasible x is given by solution v of

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

- ▶ Δx_{nt} solves second order approximation (with variable v)

$$\begin{array}{ll} \text{minimize} & \hat{f}(x+v) = f(x) + \nabla f(x)^T v + (1/2)v^T \nabla^2 f(x)v \\ \text{subject to} & A(x+v) = b \end{array}$$

- ▶ Δx_{nt} equations follow from linearizing optimality conditions

$$\nabla f(x+v) + A^T w \approx \nabla f(x) + \nabla^2 f(x)v + A^T w = 0, \quad A(x+v) = b$$

Newton decrement

- ▶ Newton decrement for equality constrained minimization is

$$\lambda(x) = (\Delta x_{\text{nt}}^T \nabla^2 f(x) \Delta x_{\text{nt}})^{1/2}$$

- ▶ gives an estimate of $f(x) - p^*$ using quadratic approximation $\hat{f}$:

$$f(x) - \inf_{Ay=b} \hat{f}(y) = \lambda(x)^2/2$$

- ▶ directional derivative in Newton direction:

$$\left. \frac{d}{dt} f(x + t \Delta x_{\text{nt}}) \right|_{t=0} = -\lambda(x)^2$$

Pf: Noting that $\nabla f(x)^T \Delta x_{\text{nt}} = -\lambda(x)^2$.

Newton's method with equality constraints

given starting point $x \in \text{dom } f$ with $Ax = b$, tolerance $\epsilon > 0$.

repeat

1. Compute the Newton step and decrement $\Delta x_{nt}, \lambda(x)$.
2. Stopping criterion. **quit** if $\lambda^2/2 \leq \epsilon$.
3. Line search. Choose step size t by backtracking line search.
4. Update. $x := x + t\Delta x_{nt}$.

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- ▶ a feasible descent method: $x^{(k)}$ feasible and $f(x^{(k+1)}) < f(x^{(k)})$
 - ▶ Δx_{nt} is feasible descent direction. (ex)
 - ▶ affine invariant (ex).

Newton's method and elimination

- ▶ reduced problem: minimize $\tilde{f}(z) = f(Fz + \hat{x})$
 - ▶ variables $z \in \mathbf{R}^{n-p}$
 - ▶ $\hat{x}$ satisfies $A\hat{x} = b$; **rank** $F = n - p$ and $AF = 0$
- ▶ (unconstrained) Newton's method for $\tilde{f}$, started at $z^{(0)}$, generates iterates $z^{(k)}$
- ▶ iterates of Newton's method with equality constraints, started at $x^{(0)} = Fz^{(0)} + \hat{x}$, are

$$x^{(k+1)} = Fz^{(k)} + \hat{x}$$

- ▶ hence, don't need separate convergence analysis

Infeasible start Newton method

Newton step at infeasible points

- ▶ with $y = (x, v)$, write optimality condition as $r(y) = 0$, where

$$r(y) = (\nabla f(x) + A^T v, Ax - b)$$

is **primal-dual residual**

- ▶ consider $x \in \text{dom } f$, $Ax \neq b$, i.e., x is infeasible
- ▶ linearizing $r(y) = 0$ gives $r(y + \Delta y) \approx r(y) + Dr(y)\Delta y = 0$
 $(\nabla f(x + \Delta x_{\text{nt}}) \approx \nabla f(x) + \nabla^2 f(x)\Delta x_{\text{nt}})$

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ \Delta v_{\text{nt}} \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T v \\ Ax - b \end{bmatrix}$$

- ▶ $(\Delta x_{\text{nt}}, \Delta v_{\text{nt}})$ is called **infeasible** or **primal-dual** Newton step at x

Infeasible start Newton method

given starting point $x \in \text{dom } f$, v , tolerance $\epsilon > 0$, $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$.
repeat
 1. Compute primal and dual Newton steps $\Delta x_{\text{nt}}, \Delta v_{\text{nt}}$.
 2. *Backtracking line search* on $\|r\|_2$.
 $t := 1$.
 while $\|r(x + t\Delta x_{\text{nt}}, v + t\Delta v_{\text{nt}})\|_2 > (1 - \alpha t)\|r(x, v)\|_2$, $t := \beta t$.
 3. *Update*. $x := x + t\Delta x_{\text{nt}}, v := v + t\Delta v_{\text{nt}}$.
until $Ax = b$ and $\|r(x, v)\|_2 \leq \epsilon$.

- ▶ not a descent method: $f(x^{(k+1)}) > f(x^{(k)})$ is possible
- ▶ directional derivative of $\|r(y)\|_2$ in direction $\Delta y = (\Delta x_{\text{nt}}, \Delta v_{\text{nt}})$ is

$$\left. \frac{d}{dt} \|r(y + t\Delta y)\|_2 \right|_{t=0} = -\|r(y)\|_2$$

Implementation

Solving KKT systems

- ▶ feasible and infeasible Newton methods require solving KKT system

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g \\ h \end{bmatrix}$$

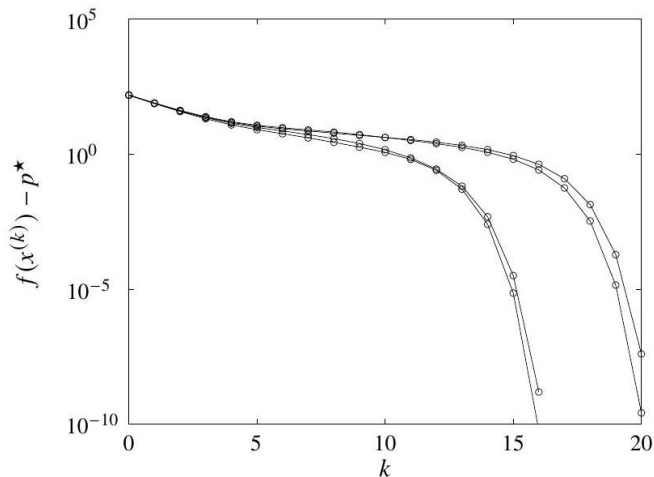
- ▶ in general, can use $\text{LDL}^\top$ factorization
- ▶ or elimination (if H nonsingular and easily inverted):
 - ▶ solve $AH^{-1}A^Tw = h - AH^{-1}g$ for w
 - ▶ $v = -H^{-1}(g + A^Tw)$

Example: Equality constrained analytic centering

- ▶ **primal problem:** minimize $-\sum_{i=1}^n \log x_i$ subject to $Ax = b$
 - ▶ **dual problem:** maximize $-b^T v + \sum_{i=1}^n \log (A^T v)_i + n$
 - ▶ recover x^* as $x_i^* = 1 / (A^T v)_i$
 - ▶ three methods to solve:
 - ▶ Newton method with equality constraints
 - ▶ Newton method applied to dual problem
 - ▶ infeasible start Newton method
- these have **different requirements for initialization**
- ▶ we'll look at an example with $A \in \mathbf{R}^{100 \times 500}$, different starting points

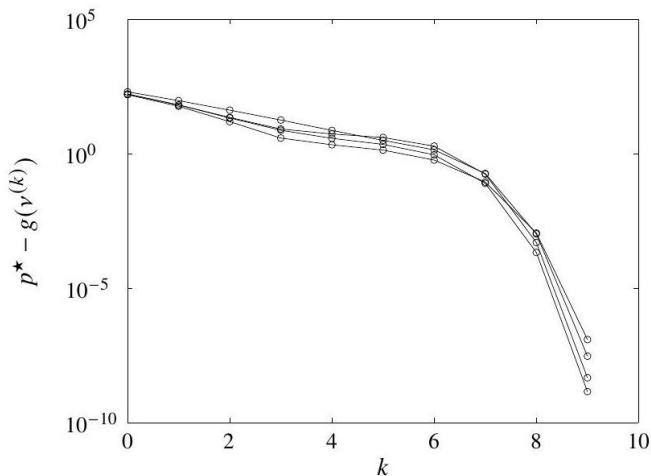
Newton's method with equality constraints

- requires $x^{(0)} > 0, Ax^{(0)} = b$



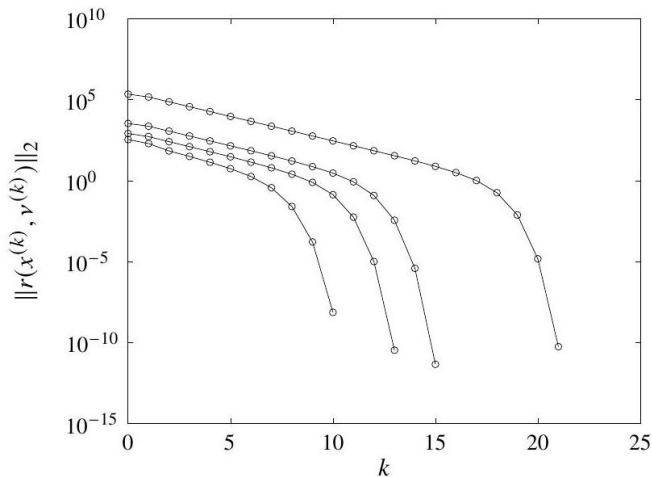
Newton method applied to dual problem

- requires $A^T v^{(0)} > 0$



Infeasible start Newton method

- requires $x^{(0)} > 0$



Complexity per iteration of three methods is identical

- ▶ for feasible Newton method, use block elimination to solve KKT system

$$\begin{bmatrix} \text{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ w \end{bmatrix} = \begin{bmatrix} \text{diag}(x)^{-1} \mathbf{1} \\ 0 \end{bmatrix}$$

reduces to solving $A \text{diag}(x)^2 A^T w = b$ (and then one can easily obtain Δx .

- ▶ for Newton system applied to dual, solve $A \text{diag}(A^T v)^{-2} A^T \Delta v = -b + A \text{diag}(A^T v)^{-1} \mathbf{1}$
- ▶ for infeasible start Newton method, use block elimination to solve KKT system

$$\begin{bmatrix} \text{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \end{bmatrix} = \begin{bmatrix} \text{diag}(x)^{-1} \mathbf{1} - A^T v \\ b - Ax \end{bmatrix}$$

reduces to solving $A \text{diag}(x)^2 A^T w = 2Ax - b$

- ▶ conclusion: in each case, solve $ADA^T w = h$ with D positive diagonal

Example: Network flow optimization

- ▶ directed graph with n arcs, $p + 1$ nodes
- ▶ x_i : flow through arc i ; ϕ_i : strictly convex flow cost function for arc i
- ▶ **incidence matrix** $\tilde{A} \in \mathbf{R}^{(p+1) \times n}$ defined as

$$\tilde{A}_{ij} = \begin{cases} 1 & \text{arc } j \text{ leaves node } i \\ -1 & \text{arc } j \text{ enters node } i \\ 0 & \text{otherwise} \end{cases}$$

- ▶ **reduced incidence matrix** $A \in \mathbf{R}^{p \times n}$ is $\tilde{A}$ with last row removed
- ▶ $\text{rank } A = p$ if graph is connected
- ▶ flow conservation is $Ax = b$, $b \in \mathbf{R}^p$ is (reduced) source vector
- ▶ **network flow optimization problem**: minimize $\sum_{i=1}^n \phi_i(x_i)$
subject to $Ax = b$

KKT system

- ▶ KKT system is

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g \\ h \end{bmatrix}$$

- ▶ $H = \mathbf{diag}(\phi_1''(x_1), \dots, \phi_n''(x_n))$, positive diagonal
- ▶ solve via elimination:

$$AH^{-1}A^T w = h - AH^{-1}g, \quad v = -H^{-1}(g + A^T w)$$

- ▶ sparsity pattern of $AH^{-1}A^T$ is given by graph connectivity

$$\begin{aligned} (AH^{-1}A^T)_{ij} \neq 0 &\iff (AA^T)_{ij} \neq 0 \\ &\iff \text{nodes } i \text{ and } j \text{ are connected by an arc} \end{aligned}$$

Analytic center of linear matrix inequality

- ▶ minimize $-\log \det X$ subject to $\operatorname{tr}(A_i X) = b_i, i = 1, \dots, p$
- ▶ optimality conditions

$$X^* > 0, \quad -(X^*)^{-1} + \sum_{j=1}^p v_j^* A_j = 0, \quad \operatorname{tr}(A_i X^*) = b_i, \quad i = 1, \dots, p$$

- ▶ Newton step ΔX at feasible X is defined by

$$X^{-1}(\Delta X)X^{-1} + \sum_{j=1}^p w_j A_j = X^{-1}, \quad \operatorname{tr}(A_i \Delta X) = 0, \quad i = 1, \dots, p$$

- ▶ follows from linear approximation
 $(X + \Delta X)^{-1} \approx X^{-1} - X^{-1}(\Delta X)X^{-1}$
- ▶ $n(n+1)/2 + p$ variables $\Delta X, w$

Solution by block elimination

- ▶ eliminate ΔX from first equation to get $\Delta X = X - \sum_{j=1}^p w_j X A_j X$
- ▶ substitute ΔX in second equation to get

$$\sum_{j=1}^p \text{tr}(A_j X A_j X) w_j = b_i, \quad i = 1, \dots, p$$

- ▶ a dense positive definite set of linear equations with variable $w \in \mathbf{R}^p$
- ▶ form and solve this set of equations to get w , then get ΔX from equation above

Flop count

- ▶ find Cholesky factor L of X $(1/3)n^3$
- ▶ form p products $L^T A_j L$ $(3/2)pn^3$
- ▶ form $p(p+1)/2$ inner products $\text{tr}((L^T A_i L)(L^T A_j L))$ to get coefficient matrix $(1/2)p^2 n^2$
- ▶ solve $p \times p$ system of equations via Cholesky factorization $(1/3)p^3$
- ▶ flop count dominated by $pn^3 + p^2 n^2$
- ▶ cf. naïve method, $(n^2 + p)^3$